

1.5 DEFINITION OF OPERATIONS RESEARCH

Any subject matter when defined to explain what exactly it is, we may find one definition. Always a definition explains what that particular subject matter is. Say for example, if a question is asked what is Boyle's law, we have a single definition to explain the same, irrespective of the language in which it is defined. But if you ask, what is Operations Research is? The answer depends on individual objective. Say for example a student may say that the Operations Research is technique use to obtain first class marks in the examination. If you ask a businessman the same question, he may say that it is the technique used for getting higher profit. Another businessman may say it is the technique used to capture higher market share and so on. Like this each individual may define in his own way depending on his objective. Each and every definition may explain one or another characteristic of Operations Research but none of them explain or give a complete picture of Operations Research. But in the academic interest some of the important definitions are discussed below.

1.6. CHARACTERISTICS OF OPERATIONS RESEARCH

After considering the objective and definitions of Operations Research, now let us try to understand what are the characteristics of Operations Research.

(a) Operations Research is an *interdisciplinary* team approach

The problems an operations research analyst face is heterogeneous in nature, involving the number of variables and constraints, which are beyond the analytical ability of one person. Hence people from various disciplines are required to understand the operations research problem, who applies their special knowledge acquired through experience to get a better view of cause and effects of the events in the problem and to get a better solution to the problem on hand. This type of team approach will reduce the risk of making wrong decisions.

Copyrighted material

12 Operations Research – Linear Programming

(b) Operations Research increases the **creative ability** of the decision maker

Operations Research provides manager mathematical tools, techniques and various models to analyse the problem on hand and to evaluate the outcomes of various alternatives and make an optimal choice. This will definitely helps him in making better and quick decisions. A manager, without the knowledge of these techniques has to make decisions by thumb rules or by guess work, which may click some times and many a time put him in trouble. Hence, a manager who uses Operations Research techniques will have a better creative ability than a manager who does not use the techniques.

(c) Operations Research is a systems approach

A business or a government organisation or a defense organisation may be considered as a system having various sub-systems. The decision made by any sub-system will have its effect on other sub-systems. Say for example, a decision taken by marketing department will have its effect on production department. When dealing with Operations Research problems, one has to consider the entire system, and characteristics or sub-systems, the inter-relationship between sub-systems and then analyse the problem, search for a suitable model and get the solution for the problem. Hence we say Operations Research is a Systems Approach.

1.8. PHASES IN SOLVING OPERATIONS RESEARCH PROBLEMS OR STEPS IN SOLVING OPERATIONS RESEARCH PROBLEMS

Any Operations Research analyst has to follow certain sequential steps to solve the problem on hand. The steps he has to follow are discussed below:

First he has to study the situation and collect all information and formulate the statement of the problem. Hence the first step is the *Formulation of the problem*. The Figure 1.3 shows the various steps to be followed.

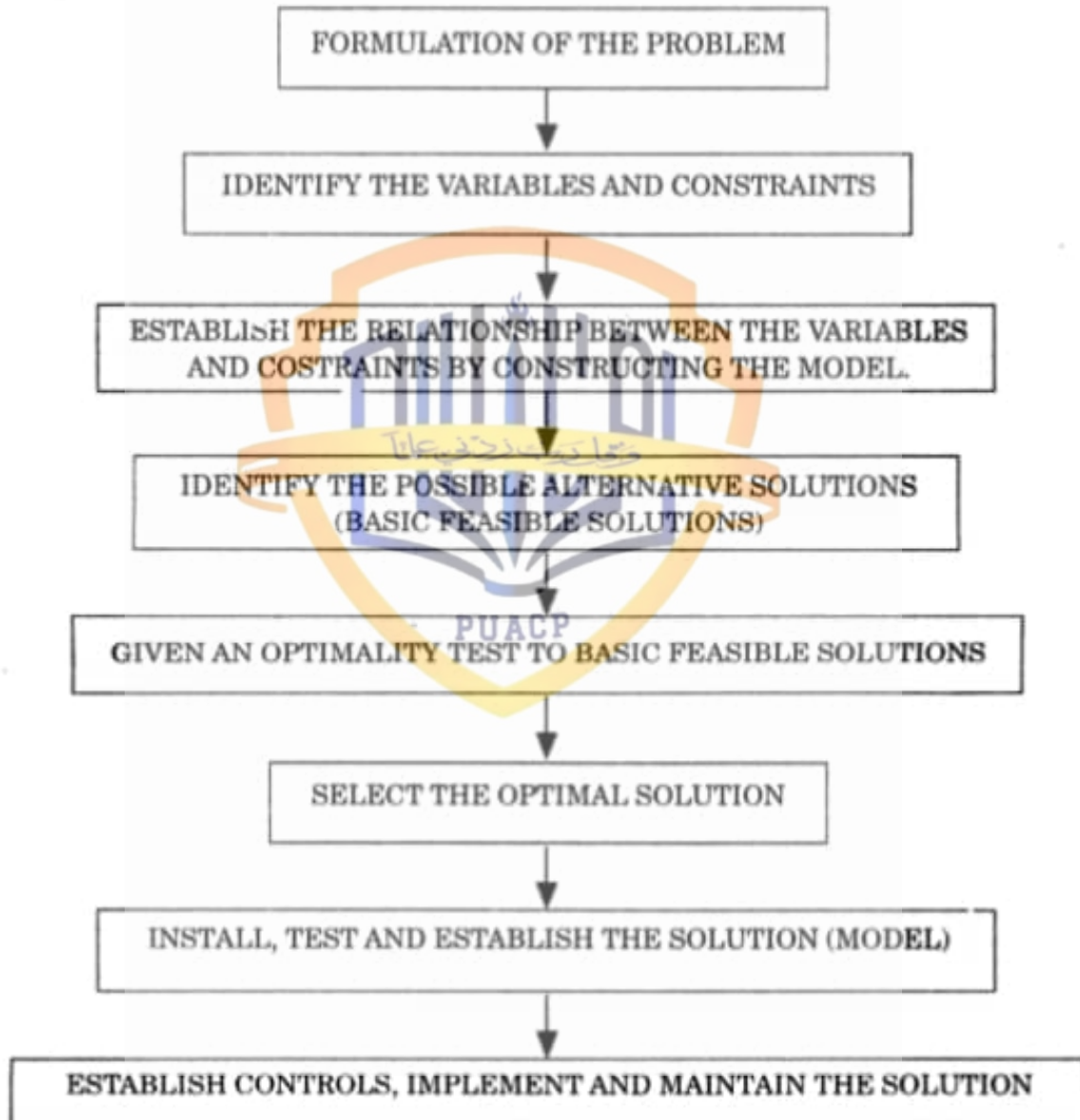


Figure 1.3. Phases of Solving Operations Research Problems.

1.10. METHODS OF SOLVING OPERATIONS RESEARCH PROBLEMS

There are three methods of solving an operations research problem. They are:

(i) Analytical method, (ii) Iterative method, (iii) The Monte-Carlo Technique.

(i) *Analytical method*: When we use mathematical techniques such as differential calculus, probability theory etc. to find the solution of a given operations research model, the method of solving is known as analytical method and the solution is known as analytical solution. Examples are problems of inventory models. This method evaluates alternative policies efficiently.

(ii) *Iterative method (Numerical Methods)*: This is trial and error method. When we have large number of variables, and we cannot use classical methods successfully, we use iterative process. First, we set a trial solution and then go on changing the solution under a given set of conditions, until no more modification is possible. The characteristics of this method is that the trial and error method used is laborious, tedious, time consuming and costly. The solution we get may not be accurate one and is approximate one. Many a time we find that after certain number of iterations, the solution cannot be improved and we have to accept it as the expected optimal solution.

PUACP

Copyrighted in

22 Operations Research – Linear Programming

(iii) *Monte-Carlo method*: This method is based on the random sampling of variable's values from a distribution of the variable. This uses sampling technique. A table of random numbers must be available to solve the problems. In fact it is a simulation process.

3.3 THE SIMPLEX METHOD

Rather than enumerating *all* the basic solutions (corner points) of the LP problem (as we did in Section 3.2), the simplex method investigates only a “select few” of these solutions. Section 3.3.1 describes the *iterative* nature of the method, and Section 3.3.2 provides the computational details of the simplex algorithm.

3.3.1 Iterative Nature of the Simplex Method

Figure 3.3 provides the **solution space** of the LP of Example 3.2-1. Normally, the simplex method starts at the **origin** (point *A*) where $x_1 = x_2 = 0$. At this starting point, the value of the objective function, z , is zero, and the logical question is whether an increase in nonbasic x_1 and/or x_2 above their current zero values can improve (increase) the value of z . We answer this question by investigating the objective function:

$$\text{Maximize } z = 2x_1 + 3x_2$$

The function shows that an increase in either x_1 or x_2 (or both) above their current zero values will *improve* the value of z . The design of the simplex method **calls** for increasing *one variable at a time*, with the selected variable being the one with the *largest rate of improvement* in z . In the present example, the value of z will increase by 2 for each unit increase in x_1 and by 3 for each unit increase in x_2 . This means that the *rate* of improvement in the value of z is 2 for x_1 and 3 for x_2 . We thus elect to increase x_2 , the variable with the largest rate of improvement. Figure 3.4 shows that the value of x_2 must be increased until corner point *B* is reached (recall that stopping short of reaching corner point *B* is not optimal because a candidate for the optimum must be a corner point). At point *B*, the simplex method will then increase the value of x_1 to reach the **improved** corner point *C*, which is the optimum. The path of the simplex algorithm is thus defined as $A \rightarrow B \rightarrow C$. Each corner point along the path is associated with an **iteration**. It is important to note that the simplex method moves alongside the **edges** of the solution space, which means that the method cannot cut across the solution space, going from *A* to *C* directly.

We need to make the transition from the graphical solution to the algebraic solution by showing how the points *A*, *B*, and *C* are represented by their basic and nonbasic variables. The following table summarizes these representations:

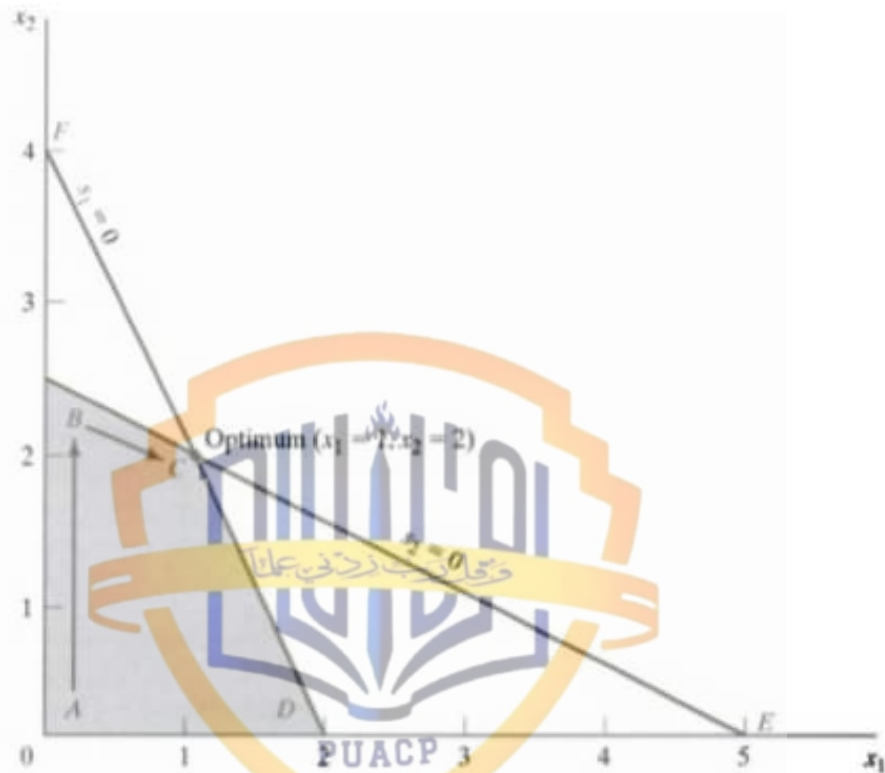


FIGURE 3.4
Iterative process of the simplex method.

Corner point	Basic variables	Nonbasic (zero) variables
A	s_1, s_2	x_1, x_2
B	s_1, x_2	x_1, s_2
C	x_1, x_2	s_1, s_2

Notice the change pattern in the basic and nonbasic variables as the solution moves along the path $A \rightarrow B \rightarrow C$. From A to B, nonbasic x_2 at A becomes basic at B and basic s_2 at A becomes nonbasic at B. In the terminology of the simplex method, we say that x_2 is the **entering variable** (because it enters the basic solution) and s_2 is the **leaving variable** (because it leaves the basic solution). In a similar manner, at point B, x_1 enters (the basic solution) and s_1 leaves, thus leading to point C.

3.5.2 Alternative Optima

When the objective function is parallel to a nonredundant **binding constraint** (i.e., a constraint that is satisfied as an equation at the optimal solution), the objective function can assume the same optimal value at more than one solution point, thus giving rise to alternative optima. The next example shows that there is an *infinite* number of such solutions. It also demonstrates the practical significance of encountering such solutions.

Copyrighted material

Example 3.5-2 (Infinite Number of Solutions)

Maximize $z = 2x_1 + 4x_2$

subject to

$$\begin{aligned}x_1 + 2x_2 &\leq 5 \\x_1 + x_2 &\leq 4 \\x_1, x_2 &\geq 0\end{aligned}$$

Figure 3.9 demonstrates how alternative optima can arise in the LP model when the objective function is parallel to a binding constraint. Any point on the line segment BC represents an alternative optimum with the same objective value $z = 10$.

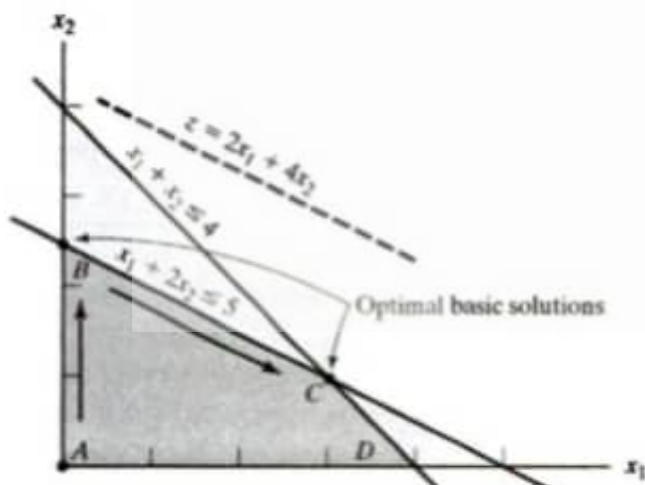


FIGURE 3.9

Alternative optima in

Example 3.5-2 (Infinite Number of Solutions)

$$\text{Maximize } z = 2x_1 + 4x_2$$

subject to

$$x_1 + 2x_2 \leq 5$$

$$x_1 + x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

Figure 3.9 demonstrates how alternative optima can arise in the LP model when the objective function is parallel to a binding constraint. Any point on the line segment BC represents an alternative optimum with the same objective value $z = 10$.

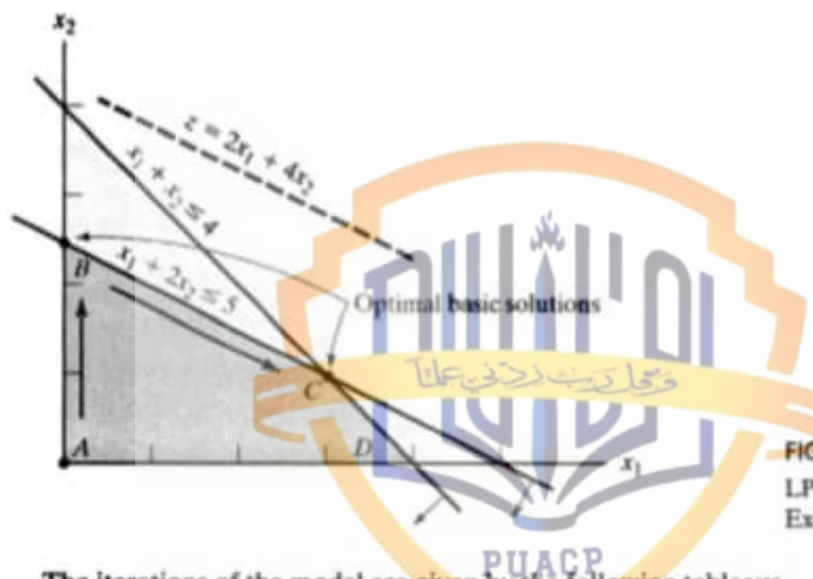


FIGURE 3.9
LP alternative optima in
Example 3.5-2

The iterations of the model are given by the following tableaus.

Iteration	Basic	x_1	x_2	x_3	x_4	Solution
0	z	-2	-4	0	0	0
x_2 enters	x_3	1	2	1	0	5
x_3 leaves	x_4	1	1	0	1	4
1 (optimum)	z	0	0	2	0	10
x_1 enters	x_2	$\frac{1}{2}$	1	$\frac{1}{2}$	0	$\frac{5}{2}$
x_4 leaves	x_4	$\frac{1}{2}$	0	$-\frac{1}{2}$	1	$\frac{3}{2}$
2	z	0	0	2	0	10
(alternative optimum)	x_2	0	1	1	-1	1
	x_1	1	0	-1	2	3

Iteration 1 gives the optimum solution $x_1 = 0$, $x_2 = \frac{5}{2}$, and $z = 10$, which coincides with point B in Figure 3.9. How do we know from this tableau that alternative optima exist? Look at the z -equation coefficients of the *nonbasic* variables in iteration 1. The coefficient of nonbasic x_1 is zero, indicating that x_1 can enter the basic solution without changing the value of z , but causing a change in the values of the variables. Iteration 2 does just that—letting x_1 enter the basic solution and forcing x_4 to leave. The new

3.5.3 Unbounded Solution

In some LP models, the values of the variables may be increased indefinitely without violating any of the constraints—meaning that the solution space is *unbounded* in at least one variable. As a result, the objective value may increase (maximization case) or decrease (minimization case) indefinitely. In this case, both the solution space and the optimum objective value are unbounded.

Unboundedness points to the possibility that the model is poorly constructed. The most likely irregularity in such models is that one or more nonredundant constraints have not been accounted for, and the parameters (constants) of some constraints may not have been estimated correctly.

The following examples show how unboundedness, in both the solution space and the objective value, can be recognized in the simplex tableau.

Example 3.5-3 (Unbounded Objective Value)

$$\text{Maximize } z = 2x_1 + x_2$$

subject to

$$x_1 - x_2 \leq 10$$

$$2x_1 \leq 40$$

$$x_1, x_2 \geq 0$$

Starting Iteration

Basic	x_1	x_2	x_3	x_4	Solution
z	-2	-1	0	0	0
x_3	1	-1	1	0	10
x_4	2	0	0	1	40

In the starting tableau, both x_1 and x_2 have negative z -equation coefficients. Hence either one can improve the solution. Because x_1 has the most negative coefficient, it is normally selected as the entering variable. However, all the constraint coefficients under x_2 (i.e., the denominators of the ratios of the feasibility condition) are negative or zero. This means that there is no leaving variable and that x_2 can be increased indefinitely without violating any of the constraints (compare with the graphical interpretation of the minimum ratio in Figure 3.6). Because each unit increase in x_2 will increase z by 1, an infinite increase in x_2 leads to an infinite increase in z . Thus, the problem has no bounded solution. This result can be seen in Figure 3.10. The solution space is unbounded in the direction of x_2 , and the value of z can be increased indefinitely.

Copyright 2012 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203

Remarks. What would have happened if we had applied the strict optimality condition that calls for x_1 to enter the solution? The answer is that a succeeding tableau would eventually have led to an entering variable with the same characteristics as x_2 . See Problem 1, Set 3.5c.

3.5.4 Infeasible Solution

LP models with inconsistent constraints have no feasible solution. This situation can never occur if *all* the constraints are of the type $\leq$ with nonnegative right-hand sides because the slacks provide a feasible solution. For other types of constraints, we use artificial variables. Although the artificial variables are penalized in the objective function to force them to zero at the optimum, this can occur only if the model has a feasible space. Otherwise, at least one artificial variable will be *positive* in the optimum iteration. From the practical standpoint, an infeasible space points to the possibility that the model is not formulated correctly.

Example 3.5-4 (Infeasible Solution Space)

Consider the following LP:

$$\text{Maximize } z = 3x_1 + 2x_2$$

subject to

$$2x_1 + x_2 \leq 2$$

$$3x_1 + 4x_2 \geq 12$$

$$x_1, x_2 \geq 0$$

Using the penalty $M = 100$ for the artificial variable R , the following tableaux provide the simplex iterations of the model.

Iteration	Basic	x_1	x_2	x_4	x_3	R	Solution
0 x_2 enters x_3 leaves	z	-303	-402	100	0	0	-1200
	x_3	2	1	0	1	0	2
	R	3	4	-1	0	1	12
1 (pseudo-optimum)	z	501	0	100	402	0	-396
	x_2	2	1	0	1	0	2
	R	-5	0	-1	-4	1	4

Copyright

Operations Research: An Introduction

Optimum iteration 1 shows that the artificial variable R is *positive* ($= 4$), which indicates that the problem is infeasible. Figure 3.11 demonstrates the infeasible solution space. By allowing the artificial variable to be positive, the simplex method, in essence, has reversed the direction of the inequality from $3x_1 + 4x_2 \geq 12$ to $3x_1 + 4x_2 \leq 12$ (can you explain how?). The result is what we may call **pseudo-optimal** solution.

SENSITIVITY ANALYSIS

In LP, the parameters (input data) of the model can change within certain limits without causing the optimum solution to change. This is referred to as *sensitivity analysis*, and will be the subject matter of this section. Later, in Chapter 4, we will study *post-optimal analysis* which deals with determining the new optimum solution resulting from making targeted changes in the input data.

In LP models, the parameters are usually not exact. With sensitivity analysis, we can ascertain the impact of this uncertainty on the quality of the optimum solution. For example, for an estimated unit profit of a product, if sensitivity analysis reveals that the optimum remains the same for a $\pm 10\%$ change in the unit profit, we can conclude that the solution is more robust than in the case where the indifference range is only $\pm 1\%$.

We will start with the more concrete graphical solution to explain the basics of sensitivity analysis. These basics will then be extended to the general LP problem using the simplex tableau results.

Graphical Sensitivity Analysis

This section demonstrates the general idea of sensitivity analysis. Two cases will be considered:

1. Sensitivity of the optimum solution to changes in the availability of the resources (right hand side of the constraints).
2. Sensitivity of the optimum solution to changes in unit profit or unit cost (coefficients of the objective function).

We will consider the two cases separately, using examples of two-variable graphical LP